LDM

$Y_t = X_t + \sigma t$ $\qquad X_t \sim F \Rightarrow Y_t \sim F_t$

$\cancel{F_{Y_t}(y) = F_{X_t}(y - \sigma t)}$ $\qquad F_t^{Y}(y) = F^{X}(y - \sigma t)$

Then $F_t(y) = \exp\left\{-(A\cancel{x})^{-\alpha}\right\} = \exp\left\{-A^{-\alpha}(y - \sigma t)^{-\alpha}\right\}$

$$f_t(y) = F_t(y) \cdot \alpha \cdot A^{-\alpha} \cdot (y - \sigma t)^{-1-\alpha}$$

Log Likelihood LDM

$$\log \mathcal{L} = \sum_{t=1}^{n} \log f_t(y)$$

$$= \sum_{t=1}^{n} -A^{-\alpha}(y_t - \sigma t)^{-\alpha} + \sum_{t=1}^{n} \log(\alpha A^{-\alpha})$$

$$+ \sum_{t=1}^{n} (-1-\alpha) \log(y_t - \sigma t)$$

$$= -\sum_{t=1}^{n} A^{-\alpha}(y_t - \sigma t)^{-\alpha} + n \log(\alpha A^{-\alpha})$$

$$- (\alpha+1) \sum_{t=1}^{n} \log(y_t - \sigma t)$$

$\sigma < \min \frac{y_t}{t}$

$\alpha A^{-\alpha} < 1 \Rightarrow \alpha < A^{\alpha} \Rightarrow \log_\alpha \alpha < \log A$

$\log \alpha^{\frac{1}{\alpha}} < \log A$

$\alpha^{\frac{1}{\alpha}} < A$

Variance of Estimators _ Kukush LDM

$$\star\ \frac{\partial \mathcal{L}}{\partial \sigma} = -\sum_{t=1}^{n} A^{-\alpha} \cdot (y_t - \sigma t)^{-\alpha-1} \cdot \alpha \cdot t + (\alpha+1) \sum_{t=1}^{n} \frac{t}{y_t - \sigma t}$$

$$\frac{\sum t\,(y_t - \sigma t)^{-\alpha-1}}{\sum t\,(y_t - \sigma t)^{-1}} = \frac{(\alpha+1)}{\alpha}$$

$$\star\star\ \frac{\partial \mathcal{L}}{\partial \sigma^2} = \sum_{t=1}^{n} A^{-\alpha} \cdot \alpha \cdot t^2 \cdot (-\alpha-1) \cdot (\underbrace{y_t - \sigma_t}_{x_t})^{-\alpha-2} + (\alpha+1) \sum_{t=1}^{n} \frac{t^2}{(y_t - \sigma t)^2}$$

$$\star\ \frac{\partial \mathcal{L}}{\partial A} = \sum_{t=1}^{n} \alpha A^{-\alpha-1} (y_t - \sigma t)^{-\alpha} - \alpha \frac{n}{A}$$

$$A = \left[\frac{\sum (y_t - \sigma t)^{-\alpha}}{n} \right]^{\frac{1}{\alpha}}$$

$$\frac{\partial \mathcal{L}}{\partial A^2} = (-\alpha-1)\, \alpha\, A^{-\alpha-2} \sum_{t=1}^{n} (y_t - \sigma t)^{-\alpha} + \alpha \frac{n}{A^2}$$

$$\frac{\partial \ell}{\partial \alpha} = \sum_{t=1}^{n} \cancel{\log(A)} A^{-\alpha} \cdot (y_t - \sigma t)^{-\alpha} \cdot \log\left[A(y_t - \sigma t)\right]$$

$$- \sum_{t=1}^{n} n \frac{\log(A) \cdot (\alpha) - 1}{\alpha} \quad \frac{\alpha \log A - 1}{\alpha}$$

$$- \sum_{t=1}^{n} \log(y_t - \sigma t)$$

$$\frac{\partial \ell}{\partial \alpha^2} = -\sum_{t=1}^{n} A^{-\alpha} (y_t - \sigma t)^{-\alpha} \left[\log A + \log(y_t - \sigma t)\right]^2$$

$$- \frac{n}{\alpha^2}$$

$$A^{-\alpha} \sum x_t^{-\alpha} \log(A x_t) + \frac{n}{\alpha}(1 - \log A) = \sum \log x_t$$

$$\sum (A x_t)^{-\alpha} \log(A x_t) + \alpha^{-1}\left[n - \log A^n\right] = \sum \log x_t$$